



“Man is what he eats”. Plant residues from dental calculus in the ancient population of Milano from Roman times to modern age

Mirko Mattia^{a,*,1}, Lucie Biehler-Gomez^a, Andrea Palamenghi^a, Deborah Nichetti^a, Giulia Caccia^a, Emanuela Sguazza^a, Danilo De Angelis^a, Paolo Maria Galimberti^b, Anna Maria Fedeli^c, Fabrizio Slavazzi^d, Cristina Cattaneo^a, Marco Caccianiga^e

^a LABANOF, Laboratorio di Antropologia e Odontologia Forense, Sezione di Medicina Legale, Dipartimento di Scienze Biomediche per la Salute, Università degli Studi di Milano, Milan, Italy

^b Servizio Beni Culturali, Fondazione IRCCS Ca' Granda Ospedale Maggiore Policlinico, Milan, Italy

^c Soprintendenza Archeologia, Belle Arti e Paesaggio per la Città Metropolitana di Milano, Milan, Italy

^d Dipartimento di Beni Culturali e Ambientali, Università degli Studi di Milano, Milan, Italy

^e Dipartimento di Bioscienze, Università degli Studi di Milano, Milan, Italy

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ABSTRACT

The study of plant residues extracted from dental calculus is a potent tool for reconstructing the past. In this study, three archaeological sites of the city of Milano (Italy) from two time periods (Roman and Modern Ages) were considered. An anthropological study (biological profile and pathologies) was performed for the 150 skeletons of the study sample and when present, residues trapped in the dental calculus were extracted. Through the observation of 883 identified starch granules, the possible changes in diet over the centuries in Milano were analysed and compared with the presence of specific signs of stress markers on the skeletons. The study showed that major cereals were a constant finding in the varied diet of the Milanese area, and that over time, legumes partially substituted the consumption of minor cereals. Interestingly, no severe nutritional deficiencies nor signs of poor diet were found in the sample. This research sheds light on the food habits of the ancient inhabitants of the city of Milano through a diachronic botanical examination of dental calculus.

1. Introduction

The study of plants used for nutritional, therapeutic or other purposes in ancient populations is an essential element for a better understanding of economy and lifestyle as well as the environmental context (Radini et al., 2017). The possibility to follow the changes of trends of these uses, such as diet, throughout centuries but within a single geographical context, can provide a better understanding of social changes within a population over time. Diet can be inferred from clues provided by the skeleton, spanning from gross pathological manifestations to trace elements and isotopes. Among these investigations, the study of microscopic debris trapped in dental calculus retrieved from

skeletal remains has the potential to provide useful insight into food consumption. Dental calculus is a mineralized bacterial plaque that builds up through multifactorial processes and naturally adheres to the teeth (Radini et al., 2017). During its formation, particles, containing food remains and residues of other materials (for example residues from the millstones used to grind cereals), can be captured and therefore protected from the chemical action of salivary amylase, so that they are preserved over time (Mickleburgh and Pagán-Jiménez, 2012). Indeed, the analysis of dental calculus could allow one to identify the plants that were used as food, to make assumptions about food preparation practices, therapeutic treatments or other habits involving plants and their contact with the oral cavity. For these reasons, dental calculus is

* Corresponding author at: LABANOF, Laboratorio di Antropologia e Odontologia Forense, Sezione di Medicina Legale, Dipartimento di Scienze Biomediche per la Salute, Università degli Studi di Milano, Via Mangiagalli 37, 20133 Milan, Italy.

E-mail address: mirko.mattia@unimi.it (M. Mattia).

¹ ORCID 0000-0002-5445-9453.

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considered as a store of biographical information on ancient populations, thus providing a novel archaeological record (Radini et al., 2017). Among food remains of plant origin that can be retrieved within dental calculus, starch has been regarded as the main source of information. Starch represents the main energy reserve of plants and the most important source of carbohydrates in the human diet (Slavin and Carlson, 2014). Starch granules can be identified at family, genus or species level, based on their size and shape providing valuable and detailed information about the diet (Henry and Piperno, 2008; Hardy et al., 2009). Other plant material that can be retrieved from dental calculus includes phytoliths (microscopic structure made of silica occurring in some plant tissues), palynomorphs (mainly pollen and spores) and other fragments that can occasionally persist. By comparing the data obtained from the study of dental calculus to historical sources, valuable information can be gained to validate the sources or implement them.

Numerous authors have studied dental calculus in various archaeological samples for the above-mentioned purposes. They provided evidence on plant consumption not only in prehistoric specimens (Mickleburgh and Pagán-Jiménez, 2012; Buckley et al., 2014; Tromp and Dudgeon, 2015) but also in more recent samples, such as Roman and Medieval time periods (Belcastro et al., 2007; Hardy et al., 2009; Radini, Nikita and Shillito, 2016; Gismondi et al., 2018). However, they have often focused on single time periods or in different places, making it difficult to compare how consumption evolved in different time periods within the same population.

This paper presents a diachronic pilot study on the comparative analyses of plant material trapped in calculus within these two periods in the city of Milano. We considered any identifiable plant remains, keeping the main focus on starch grains but recording any other plant evidence taxonomically attributable. Results were then correlated with data from the anthropological analysis of the skeletons. To the best of our knowledge, this is the first investigation focusing on the changes in dietary habits of an urban population, comparing Roman (3rd century CE) and Modern (17th century CE) periods within Milano, as well as two different social contexts from the 17th century: a mass grave for “normal” citizens and a burial site reserved for patients who died in the city hospital. This is part of a larger project called “MiAntropo” that aims to reconstruct the life of the city of Milano throughout time. It is an ongoing anthropological, archaeological and historical project that strives to create a multidisciplinary archive for diachronically tracing the history of Milano through the skeletal remains of people who lived in the city and whose life is often not reported by historical sources.

2. Materials and methods

For this study, three archaeological sites of the city of Milano (Italy) and two time periods (Roman Age and Modern Age) were considered. In this pilot study, 50 skeletons from each site were selected randomly among the skeletons with the best state of preservation. Only adolescents and adults (13–20 years old) were considered.

The sites are:

- Milano Cattolica: necropolis of the Roman-Imperial age (3rd–4th century CE).
- Milano Sabotino: modern age site, dated to the first half of the 17th century CE. The skeletal remains were found in a common grave, divided into groups, in a context that suggests a hasty burial due to an epidemic (Caruso et al., 2013).
- Milano Ca' Granda: hospital burial in use, according to archival sources, in the second half of the 17th century CE; however, recent bioarchaeological analyses do not exclude a subsequent use of the sepulchral spaces. A huge number (an estimated 2 million bones) of highly commingled skeletal remains were found in the underground crypts, below the most important hospital of Milano between the 15th and the 19th century, which was dedicated to the care of the poorest citizens in the city (Cosmacini, 1999).

2.1. Anthropological analyses

All 150 individuals were examined: sex, age, ancestry, stature and pathology were established according to common standards (Buikstra, 2019a; Beck, 1995). In particular, the following pathological signs and stress markers were scored for each individual: degenerative and occupational markers (such as arthritis and enthesopathies), periostitis, *cribra orbitalia* (Fig. 1), enamel hypoplasia (Fig. 2) and classical signs of rickets and scurvy (Brickley, 2018; Brickley and Mays, 2019; Kinaston et al., 2019).

As regards dental analysis, a specific chart was filled in to document the presence or absence (antemortem or postmortem) of each dental element and dental pathology (caries, abscess, hypoplasia, calculus, wear, Kinaston et al., 2019). As for dental calculus, its presence was assessed following a score-based classification (Dobney and Brothwell, 1987). Only 55 individuals from the selected sample of 150 individuals presented a score of 1 or more relative to dental calculus (a score of 1 being the minimum requirement for inclusion in the subset)

2.2. Plant remain analyses

From each tooth that presents dental calculus, this was collected, treated separately and stored in an Eppendorf tube. Pictures of the dental plaque were taken before and after the collection of the dental calculus. Calculus quantities varied among the samples, ranging from 0.0001 g to 0.488 g.

Contamination tests were not conducted at the excavation sites; however, several tests were performed in order to identify potential contamination throughout the subsequent laboratory work. In particular, control samples were performed to identify potential contaminants from the working and storage environment. The plant remains analyses were performed in a bioscience laboratory that provides strict rules for cleaning of working areas and equipment (where contamination controls were regularly carried out on workspaces, instruments and supplies); furthermore blanks (negative control samples consisting of sterile, filtered water) were created for each site and method during decalcification in order to detect potential areas of contamination during processing. There were no starch granules recovered from these blanks. This suggests that there was little to no contamination throughout the sample processing activities.

After dental calculus collection, in order to eliminate any past and modern environmental contaminant from calculus surface, the samples underwent consecutive washing cycles of 10 min each (Tavarone et al., 2018) in a sterile Eppendorf with ultra-pure water and centrifugation at 5000 rpm for 10 min.

Each sample was weighted and then suspended with 1.5 mL of HCl (3 M). This dissolve nearly all of the calculus within one week (Henry

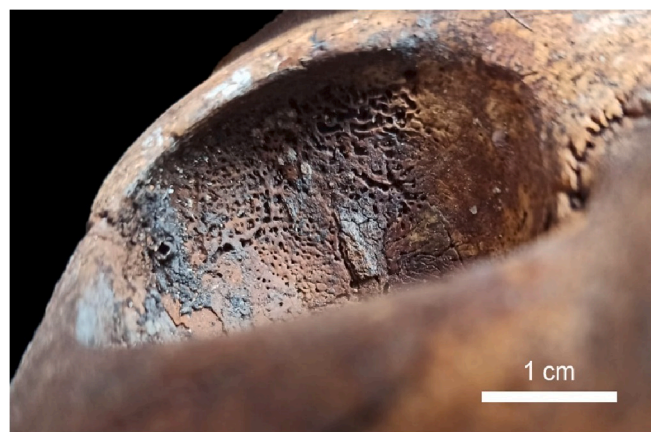


Fig. 1. Cribra orbitalia from MI CG O 6.



Fig. 2. Enamel hypoplasia from T 2 GRUPPO 11 IND H.

and Piperno, 2008; Hardy et al., 2009; Wesolowski et al., 2010; Henry, Brooks and Piperno, 2011; Mickleburgh and Pagán-Jiménez, 2012). The resulting pellet underwent two consecutive washings with distilled water and then centrifugation at 5000 rpm for 10 min.

The samples were placed on a microscope slide and coloured with Lugol in order to perform microscopic analysis. The samples were then observed with an optical microscope (Leica – Leitz DMRD) under different magnifications and captured with Leica MC170 HD camera. Taxonomic identification at the lowest possible level was performed considering morphology and dimension of the granules and by comparison with a reference collection held at the laboratory that included plants currently used for nutrition (Table 1; Table 2); the reference collection includes both raw samples and samples boiled for 20 min to take into account changes in starch grain morphology due to cooking.

The species were then regrouped into macro categories following their taxonomic affinity, their use and diffusion and their reciprocal.

3. Results

3.1. Anthropological results of the sample

Table 3 summarises the biological profile of the original sample of 150 individuals. In addition, pathological indicators, stress markers and starch results are reported. As regards population affinity, all the

individuals were of European ancestry.

As shown in Table 3, only 55 individuals presented dental calculus and were hence selected in the study. In this subset, estimation of sex was available for 43 of the 55 skeletons under examination. Age-at-death estimation showed that the sample was composed of 48 adult individuals and 6 subadults; only one individual remained undetermined. As for the pathological analysis, 21 individuals with stress markers were observed.

3.2. Plant remains

Overall, 1019 starch grains were observed. 87% ($n = 883$) of these grains were successfully identified at least at the macro category level, 10% were not determined because of the lack of discriminant morphological features, and the rest (3%) included grains that were broken, deformed and hidden from extraneous material, therefore, their identification was not possible. However, the unambiguous identification at the species level was not always possible, as a certain degree of uncertainty and traits overlapping commonly occurred, particularly within the cereal macro category and specifically between barley and the *Triticum* species. For this reason, the obtained identification at the species level should always be considered only as the most likely, and comparisons were performed at the macro category level. Table 4 summarises the number of identified starch grains in each site and the total number of each species. In general, cereals represented the most frequent finding. In descending order, cereals were mainly represented by barley, rye, sorghum, spelt, oat and common wheat. Rice, corn, millet and durum wheat were less frequently found. Legumes were mainly represented by lentil, broad bean, bean (possibly including cowpea, *Vigna unguiculata*), chickpea and pea.

3.2.1. Milano Ca' Granda

In Milano Ca' Granda (430 starch grains identified), high frequencies of cereals, such as barley, rye and spelt, were observed (Fig. 3). The least represented cereals were corn, rice, millet and durum wheat, along with buckwheat. The most frequent legumes were lentils and broad bean. Pea, chestnut and potato were rarely observed, while wild pea was completely absent.

3.2.2. Milano Sabotino

In Milano Sabotino (337 starch grains identified), cereals were mainly represented by sorghum, followed by barley and rye (Fig. 3). Rice, corn, millet, and buckwheat were present, though less represented. Lentils were the most represented legume, while broad bean, bean and chickpea occurred less frequently. Buckwheat was found whereas durum wheat, potato, chestnut and wild pea were never observed.

3.2.3. Milano Cattolica

The three individuals from Cattolica (114 starch grains identified) showed high frequencies of cereals, such as wheat, barley, rye and spelt. All species of minor cereals considered in this study were detected. Legumes were represented mainly by lentils.

3.2.4. Other plant remains

Other plant elements such as pollen and spores were occasionally recovered: an interesting and unexpected finding was field horsetail (*Equisetum arvense*) spores, observed in all the investigated sites.

In addition, microscopic fragments of xylem and unidentified fungal spores, were observed in two individuals from the Cattolica site.

4. Discussion

Starch granules are considered an effective means for the reconstruction of ancient human diet. Extraction and taxonomic identification of starch grains from the dental calculus of archaeological skeletal remains can provide valuable insight into nutritional habits of past

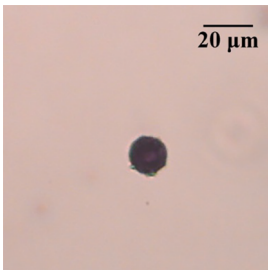
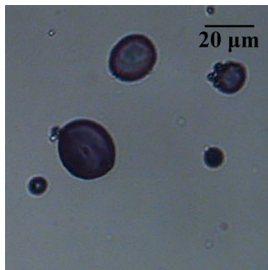
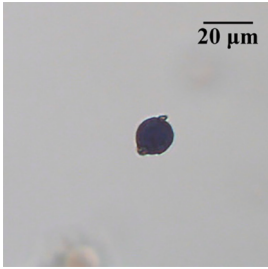
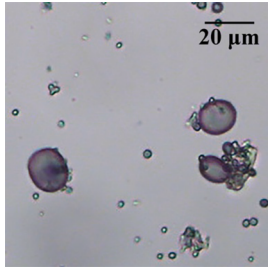
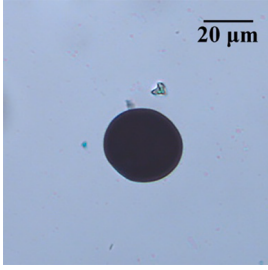
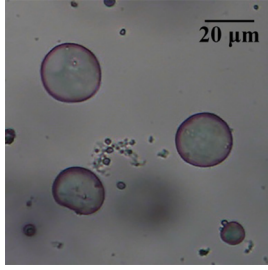
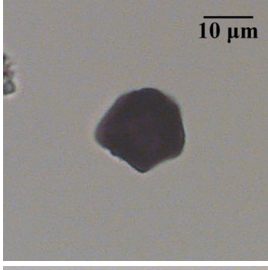
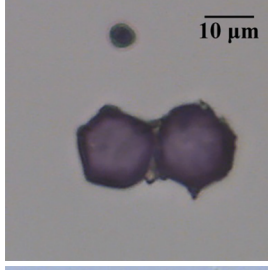
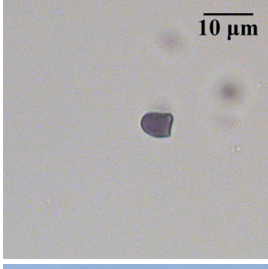
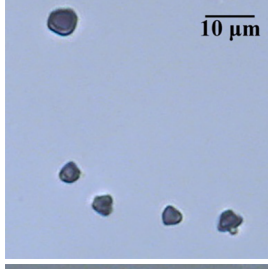
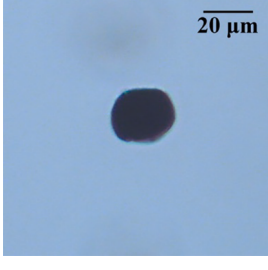
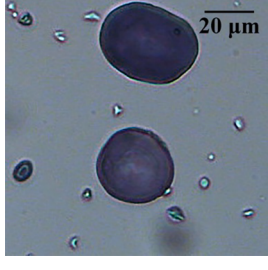
Table 1

Taxonomical subdivision of plant remains in macro categories.

Species	Common name	Group
<i>Triticum</i> spp	Spelt, Emmer	CEREALS
<i>Hordeum</i> spp	Barley	
<i>Secale cereale</i>	Rye	
<i>Zea mais</i>	Corn	MINOR CEREALS
<i>Oryza sativa</i>	Rice	
<i>Triticum durum</i>	Durum wheat	
<i>Triticum aestivum</i>	Common wheat	
<i>Avena sativa</i>	Oat	
<i>Panicum miliaceum</i>	Millet	
<i>Sorghum bicolor</i>	Sorghum	
<i>Lens culinaris</i>	Lentil	LEGUMES
<i>Vicia faba</i>	Broad bean	
<i>Cicer arietinum</i>	Chickpea	
<i>Pisum sativum</i>	Pea	
<i>Phaseolus vulgaris</i>	Bean	OTHERS
<i>Castanea sativa</i>	Chestnut	
<i>Solanum tuberosum</i>	Potato	
<i>Fagopyrum esculentum</i>	Buck wheat	

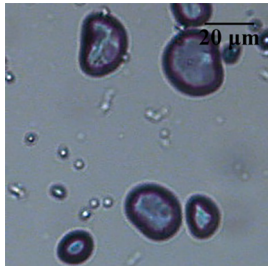
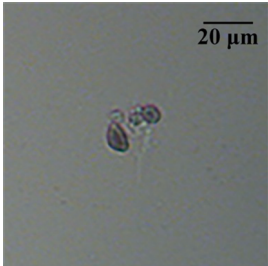
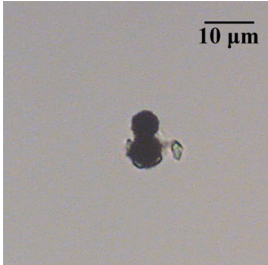
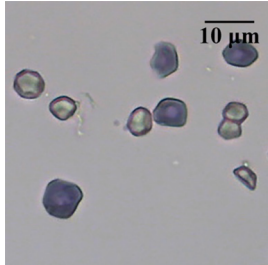
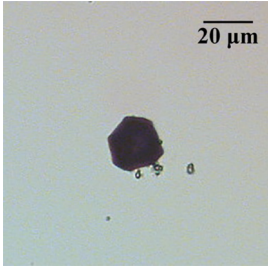
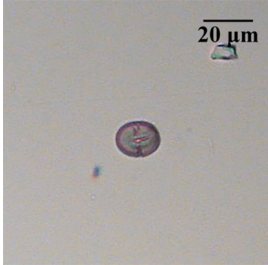
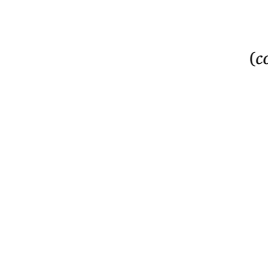
Table 2

Reference collection used for the identification of the starch granules extracted from the dental calculus of the skeletal remains, in the three sites.

STARCH GRAINS	Images from the samples of the three investigated sites	Images from the comparison collection (raw material)
SPELT		
BARLEY		
RYE		
CORN		
RICE		
DURUM WHEAT		
COMMON WHEAT		

(continued on next page)

Table 2 (continued)

STARCH GRAINS	Images from the samples of the three investigated sites	Images from the comparison collection (raw material)
OAT		
		
MILLET		
		
SORGHUM		
		
LENTIL		
BROAD BEAN		
CHICKPEA		

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Table 2 (continued)

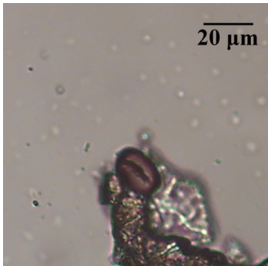
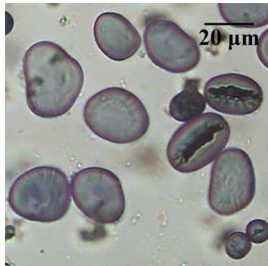
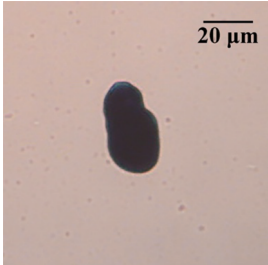
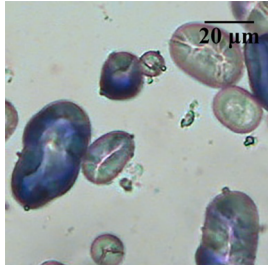
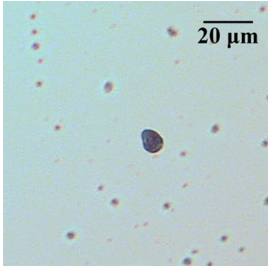
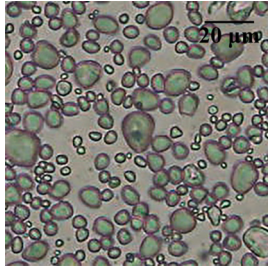
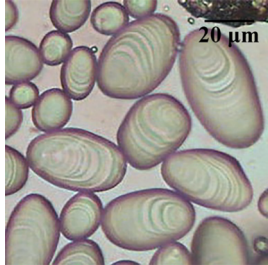
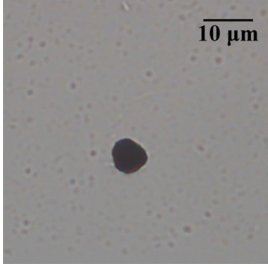
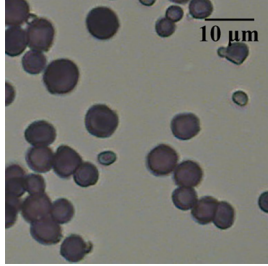
STARCH GRAINS	Images from the samples of the three investigated sites	Images from the comparison collection (raw material)
PEA		
		
BEAN		
		
CHESTNUT		
		
POTATO		
		
BUCK WHEAT		

Table 3
Anthropological and plant remain results.

SITE	ID	Time period	SEX	AGE	Stature	Pathology	Plant remains	Other
MILANO CATTOLICA	T 833 US 833	III - V c. AD	Male	Adult ND	ND	Dental hypoplasia, dental wear, arthritis and enthesopathies		
	US 1112	III - V c. AD	ND	31–45 y	ND	None		
	US 1231	III - V c. AD	Male	31–45 y	ND	None		
	US 1232	III - V c. AD	Male	31–45 y	ND	Dental wear, caries, arthritis, enthesopathies		
	US 1354	III - V c. AD	Female	31–45 y	ND	Arthritis		
	US 1362	III - V c. AD	Male	13–19 y	ND	Dental hypoplasia		
	US 1369	III - V c. AD	Male	13–19 y	ND	Dental hypoplasia and arthritis		
	US 1431B	III - V c. AD	Male	31–45 y	ND	Dental hypoplasia and dental wear		
	US 1431C	III - V c. AD	Male	20–30 y	ND	Dental hypoplasia		
	US 1534	III - V c. AD	Male	31–45 y	ND	Arthritis and enthesopathies		
	US 1540	III - V c. AD	Male	20–30 y	ND	None		
	US 1541	III - V c. AD	Male	20–30 y	ND	Dental wear, enthesopathies and trauma		
	US 1546	III - V c. AD	Male	20–30 y	ND	Arthritis and enthesopathies		
	US 1589 = 1960	III - V c. AD	ND	20–30 y	ND	None		
	US 1604	III - V c. AD	Female	20–30 y	177 cm	Dental wear, caries and arthritis		
	US 1692	III - V c. AD	Male	31–45 y	167 cm	Dental wear, arthritis and trauma		
	US 1777	III - V c. AD	Female	31–45 y	ND	Dental wear and arthritis		
	US 1804	III - V c. AD	Male	46–60 y	171 cm	Arthritis and enthesopathies		
	US 1805	III - V c. AD	Male	31–45 y	166 cm	Dental wear and arthritis		
	T 1879 US 1878	III - V c. AD	Male	31–45 y	166 cm	Arthritis, enthesopathies, trauma and inflammations		
	US 1983	III - V c. AD	Male	31–45 y	157 cm	None		
	US 2085	III - V c. AD	Male	46–60 y	168 cm	Arthritis, enthesopathies and trauma		
	2113	III - V c. AD	Male	46–60 y	171 cm	Arthritis and trauma		
	2141	III - V c. AD	Male	20–30 y	173 cm	Dental hypoplasia, cribra orbitalia and arthritis		
	US 2142	III - V c. AD	Male	20–30 y	ND	Dental hypoplasia		
	US 2283	III - V c. AD	Female	31–45 y	ND	Dental wear		
	US 2380	III - V c. AD	Female	31–45 y	160 cm	Enthesopathies		
	US 2380	III - V c. AD	Male	20–30 y	ND	Dental wear and caries		
	US 2441	III - V c. AD	Female	31–45 y	158 cm	Arthritis		
	US 2626	III - V c. AD	Male	31–45 y	ND	None		
	US 2672C	III - V c. AD	Female	31–45 y	ND	None		
	US 2787	III - V c. AD	Male	46–60 y	ND	Arthritis, enthesopathies and inflammations		
	US 2832	III - V c. AD	Male	31–45 y	173 cm	Trauma		
	US 2842	III - V c. AD	Female	20–30 y	ND	Dental wear		
	US 2877	III - V c. AD	Male	31–45 y	ND	None		
	US 2941	III - V c. AD	Female	31–45 y	150 cm	None		
	US 2943		Female		ND	None		

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Table 3 (continued)

SITE	ID	Time period	SEX	AGE	Stature	Pathology	Plant remains	Other	
MILANO SABOTINO	US 2948 A	III - V c. AD	Female	13–19 y	ND	None	Oat (1), Spelt (2), Broad bean (1), Lentil (3), Sorghum (11), Rye (2), Common wheat (1), Millet (3)	Horsetail (3)	
	US 2952	III - V c. AD		31–45 y					
	T 3110 US 3112	III - V c. AD	ND	Adult ND	176 cm	Arthritis and enthesopathies			
	US 4027	III - V c. AD	Male	31–45 y	ND	Dental calculus, dental wear, caries, arthritis and enthesopathies			
	T 4066 US 4068	III - V c. AD	Male	31–45 y	167 cm	Dental hypoplasia, dental wear, trauma, arthritis and enthesopathies			
	T 4193 US 4191	III - V c. AD	Male	46–60 y	170 cm	Arthritis and enthesopathies			
	T 4706 US 5273	III - V c. AD	Male	46–60 y	165 cm	Arthritis, enthesopathies and inflammations			
	T 4950 US 4949	III - V c. AD	ND	Adult ND	169 cm	Trauma			
	T 5284 US 5283	III - V c. AD	Female	Adult ND	159 cm	Dental wear, arthritis, trauma and enthesopathies			
	US 5398	III - V c. AD	Male	31–45 y	ND	Arthritis and trauma			
	MI SA	III - V c. AD	Male	46–60 y	168 cm	Dental calculus, dental wear, arthritis and trauma			Oat (12), Barley (13), Common wheat (4), Millet (1), Lentil (1), Spelt (1)
	MI SE	III - V c. AD	Male	20–30 y	180 cm	Tuberculosis, Harris lines and trauma			
	MI SI	III - V c. AD	Male	20–30 y	180 cm	Dental calculus, cribra orbitalia, trauma and inflammations	Oat (7), Spelt (9), Barley (17), Lentil (6), Pae (2), Millet (1), Common wheat (2), Chickpea (1), Chestnut (1)		
	T 2 GRUPPO 1 IND B	XVII c. AD	Female	46–60 y	ND	Trauma			
	T 2 GRUPPO 1 IND F	XVII c. AD	Female	46–60 y	160 cm	Arthritis and enthesopathies			
	T 2 GRUPPO 2 IND A	XVII c. AD	Female	31–45 y	167 cm	Trauma and inflammations			
	T 2 GRUPPO 2 IND B	XVII c. AD	Male	20–30 y	175 cm	Partial edentulism, cribra orbitalia and arthritis			
	T 2 GRUPPO 2 IND C	XVII c. AD	Female	20–30 y	163 cm	Partial edentulism, dental hypoplasia, cribra orbitalia and enthesopathies			
	T 2 GRUPPO 2 IND D	XVII c. AD	Female	31–45 y	163 cm	Partial edentulism and arthritis			
	T 2 GRUPPO 2 IND E	XVII c. AD	Male	46–60 y	169 cm	Partial edentulism, dental calculus, dental wear, dental hypoplasia, caries, arthritis and enthesopathies	Lentil (3), Sorghum (1), Millet (1)		
	T 2 GRUPPO 2 IND F	XVII c. AD	Male	46–60 y	160 cm	Dental wear, dental hypoplasia, caries and arthritis	Bean (5), Oat (1), Spelt (6), Broad bean (1), Lentil (6), Sorghum (2), Barley (3), Rye (5), Common wheat (1)		
	T 2 GRUPPO 2 IND G	XVII c. AD	Male	20–30 y	ND	Dental wear, dental hypoplasia, trauma, arthritis	Oat (2), Spelt (5), Broad bean (1), Lentil (5), Sorghum (1), Barley (10), Rye (3), Common wheat (3), Rice (1), Buck wheat (1)		Horsetail (3)
	T 2 GRUPPO 2 IND I	XVII c. AD	Female	31–45 y	ND	Partial edentulism, arthritis and enthesopathies			
	T 2 GRUPPO 2 IND L	XVII c. AD	Male	31–45 y	179 cm	Dental calculus, dental wear, dental hypoplasia, caries, cribra orbitalia, arthritis and enthesopathies	Spelt (5), Broad bean (1), Lentil (1), Sorghum (7), Barley (4), Rye (16), Corn (4), Buck wheat (1)		
	T 2 GRUPPO 3 IND C	XVII c. AD	Male	46–60 y	163 cm	Partial edentulism, dental wear, caries and arthritis			
	T 2 GRUPPO 3 IND D	XVII c. AD	Male	20–30 y	ND	Arthritis and enthesopathies			
	T 2 GRUPPO 3 IND E	XVII c. AD	Male	46–60 y	ND	Tuberculosis			
	T 2 GRUPPO 3 IND N	XVII c. AD	Male	46–60 y	170 cm	Partial edentulism, dental wear, caries, arthritis, enthesopathies			
	T 2 GRUPPO 4 IND B	XVII c. AD	Female	> 60 y	160 cm	Arthritis			
	T 2 GRUPPO 4 IND C	XVII c. AD	Male	46–60 y	165 cm	Partial edentulism, dental calculus, dental wear, caries and arthritis	Oat (1), Chickpea (1), Spelt (4), Broad bean (1), Lentil (7), Sorghum (2), Barley (6), Rye (3)		
	T 2 GRUPPO 4 IND E	XVII c. AD	Female	20–30 y	162 cm	Partial edentulism, dental wear, caries, abscess and other inflammations			
	T 2 GRUPPO 4 IND F	XVII c. AD	Female	31–45 y	157 cm	Partial edentulism and arthritis			
				Female	ND				

(continued on next page)

Table 3 (continued)

SITE	ID	Time period	SEX	AGE	Stature	Pathology	Plant remains	Other
MILANO CA' GRANDA	T 2 GRUPPO 4 IND G	XVII c. AD		13–19 y		Dental calculus, dental wear, dental hypoplasia and caries	Chickpea (1), Lentil (1), Sorghum (1), Rice (1)	
	T 2 GRUPPO 4 IND H	XVII c. AD	Male	31–45 y	ND	Partial edentulism, dental wear, caries and arthritis		
	T 2 GRUPPO 4 IND L	XVII c. AD	Male	31–45 y	167 cm	Partial edentulism, dental calculus, dental wear, dental hypoplasia, caries and arthritis	Oat (1), Spelt (1), Broad bean (1), Lentil (2), Sorghum (1), Rye (2)	
	T 2 GRUPPO 4 IND M	XVII c. AD	Male	20–30 y	171 cm	Dental wear, dental hypoplasia, caries and arthritis		
	T 2 GRUPPO 4 IND N	XVII c. AD	Female	13–19 y	ND	Dental calculus, dental hypoplasia and caries	Bean (4), Chickpea (1), Broad bean (2)	
	T 2 GRUPPO 4 IND R	XVII c. AD	Male	46–60 y	167 cm	Dental wear, dental hypoplasia, caries and arthritis		
	T 2 GRUPPO 5 IND C	XVII c. AD	Male	46–60 y	ND	Partial edentulism, dental calculus, dental wear, caries and arthritis	Bean (1), Oat (1), Spelt (1), Broad bean (1), Lentil (5), Sorghum (20), Barley (5), Rye (1)	
	T 2 GRUPPO 5 IND E	XVII c. AD	Male	46–60 y	ND	Partial edentulism, dental calculus, dental wear, caries, arthritis and inflammations	Bean (2), Oat (1), Chickpea (1), Lentil (1), Sorghum (1), Barley (1), Rye (6)	
	T 2 GRUPPO 5 IND F	XVII c. AD	Male	Adult ND	ND	Partial edentulism, dental calculus, dental wear, dental hypoplasia, caries and abscess	Bean (1), Oat (1), Chickpea (1), Spelt (7), Lentil (3), Sorghum (3), Barley (7), Rye (1), Common wheat (1), Rice (1)	
	T 2 GRUPPO 5 IND G	XVII c. AD	Female	46–60 y	ND	Partial edentulism and arthritis		
	T 2 GRUPPO 6 IND D	XVII c. AD	Male	46–60 y	172 cm	ND		
	T 2 GRUPPO 6 IND E	XVII c. AD	Female	46–60 y	ND	Arthritis and enthesopathies		
	T 2 GRUPPO 6 IND F	XVII c. AD	Female	20–30 y	155 cm	Dental calculus, dental hypoplasia, caries and trauma	Sorghum (1), Rye (1)	
	T 2 GRUPPO 6 IND L	XVII c. AD	Female	13–19 y	160 cm	Dental calculus, dental hypoplasia, caries and arthritis	Oat (1), Sorghum (2), Rye (1), Common wheat (1)	
	T 2 GRUPPO 7 IND E	XVII c. AD	Female	31–45 y	ND	Arthritis		
	T 2 GRUPPO 7 IND F	XVII c. AD	Male	31–45 y	ND	Partial edentulism and dental wear		
	T 2 GRUPPO 9 IND D	XVII c. AD	Female	46–60 y	ND	ND		
	T 2 GRUPPO 10 IND A	XVII c. AD	Female	13–19 y	ND	ND		
	T 2 GRUPPO 10 IND B	XVII c. AD	Female	46–60 y	ND	Dental calculus, dental wear and caries	Broad bean (1), Lentil (1)	
	T 2 GRUPPO 10 IND D	XVII c. AD	Female	46–60 y	155 cm	Partial edentulism and arthritis		
	T 2 GRUPPO 10 IND E	XVII c. AD	Male	31–45 y	ND	Partial edentulism, dental calculus, dental wear, dental hypoplasia and caries	Bean (1), Oat (1), Chickpea (1), Lentil (7), Sorghum (5), Barley (3), Rye (1)	Horsetail (1)
	T 2 GRUPPO 10 IND F	XVII c. AD	Female	46–60 y	ND	Partial edentulism		
	T 2 GRUPPO 10 IND H	XVII c. AD	Male	46–60 y	171 cm	Partial edentulism, dental calculus, dental wear, dental hypoplasia, caries, abscess, trauma and arthritis	Bean (1), Broad bean (4), Sorghum (2), Barley (1), Rye (1), Common wheat (1)	
	T 2 GRUPPO 10 IND N	XVII c. AD	Male	46–60 y	167 cm	Trauma and arthritis		
	T 2 GRUPPO 11 IND AA	XVII c. AD	Male	46–60 y	172 cm	Arthritis		
	T 2 GRUPPO 11 IND F	XVII c. AD	Male	46–60 y	ND	Dental calculus, trauma, arthritis and enthesopathies	Lentil (2), Sorghum (3), Barley (4), Rye (3), Corn (1)	
	T 2 GRUPPO 11 IND H	XVII c. AD	Female	46–60 y	ND	Dental hypoplasia and caries		
	T 2 GRUPPO 11 IND I	XVII c. AD	Male	31–45 y	157 cm	Dental calculus, dental hypoplasia and caries	Broad bean (2), Lentil (4), Sorghum (5), Barley (2), Rye (1), Rice (2)	
	T 2 GRUPPO 11 IND N	XVII c. AD	Female	20–30 y	155 cm	Arthritis		
	T 2 GRUPPO 12 IND A	XVII c. AD	Male	31–45 y	ND	Dental calculus, dental hypoplasia, caries and arthritis	Oat (1), Chickpea (1), Spelt (1), Lentil (3), Sorghum (2), Barley (4), Rye (2), Common wheat (1), Millet (1)	Horsetail (2)
	T 2 GRUPPO 15 IND M	XVII c. AD	Male	46–60 y	ND	Dental calculus, dental wear, dental hypoplasia and arthritis	Chickpea (4), Spelt (2), Broad bean (1), Lentil (3), Sorghum (2), Barley (7), Rye (10), Common wheat (1)	
	MI CG O 1	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Spelt (1), Lentil (3), Sorghum (5), Barley (3), Rye (1), Buck wheat (3), Millet (4)	
	MI CG O 2	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Spelt (2), Broad bean (4), Lentil (4), Sorghum (2), Barley (17), Rye (8), Common wheat (4), Durum wheat (1)	Horsetail (2)

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Table 3 (continued)

SITE	ID	Time period	SEX	AGE	Stature	Pathology	Plant remains	Other
	MI CG O 3	XVII c. AD	Female	Adult ND	ND	Dental calculus, dental wear and cribra orbitalia	Bean (3), Oat (2), Spelt (14), Lentil (9), Sorghum (2), Barley (7), Rye (22), Common wheat (6), Rice (1), Corn (4), Chestnut (1)	
	MI CG O 4	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Bean (1), Chickpea (1), Spelt (2), Broad bean (3), Lentil (4), Sorghum (5), Barley (13), Rye (5), Common wheat (4), Potato (1)	Horsetail (2)
	MI CG O 5	XVII c. AD	Female	Adult ND	ND	Dental calculus and dental wear	Bean (1), Spelt (6), Broad bean (6), Barley (7), Rye (1)	
	MI CG O 6	XVII c. AD	Male	Adult ND	ND	Dental calculus, dental wear and cribra orbitalia	Bean (1), Oat (2), Barley (2), Corn (1), Millet (1), Potato (3)	
	MI CG O 7	XVII c. AD	ND	Adult ND	ND	Partial edentulism, dental calculus and dental wear	Chickpea (1), Spelt (1), Lentil (1), Common wheat (1)	
	MI CG O 8	XVII c. AD	ND	Adult ND	ND	Dental calculus and dental wear	Spelt (1), Lentil (3), Corn (1), Pea (1)	
	MI CG A 9	XVII c. AD	ND	13–19 y	ND	Dental calculus and dental wear	Oat (1), Corn (1)	
	MI CG A 10	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Bean (2), Chickpea (1)	
	MI CG A 11	XVII c. AD	Male	Adult ND	ND	Partial edentulism, dental calculus, dental wear and dental hypoplasia	Bean (1), Oat (6), Spelt (1), Broad bean (3), Lentil (1), Barley (1), Rye (6), Rice (8), Potato (8)	
	MI CG D1 12	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Oat (1), Broad bean (3), Sorghum (2), Rye (1)	
	MI CG D1 13	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Bean (1), Oat (1), Chickpea (1), Lentil (1), Corn (1)	
	MI CG D1 14	XVII c. AD	ND	Adult ND	ND	Dental calculus and dental wear	Bean (1), Oat (1), Chickpea (1), Rye (1)	
	MI CG D2 15	XVII c. AD	Male	Adult ND	ND	Partial edentulism, dental calculus and dental wear	Oat (2), Broad bean (2), Lentil (1), Sorghum (1), Barley (1), Rice (1)	Horsetail (1)
	MI CG D2 16	XVII c. AD	ND	Adult ND	ND	Dental calculus and dental wear	Chickpea (3), Broad bean (1), Sorghum (1), Common wheat (1), Potato (2)	
	MI CG D2 17	XVII c. AD	Male	ND	ND	Dental calculus, dental wear and caries	Oat (1)	
	MI CG F 18	XVII c. AD	ND	Adult ND	ND	Dental calculus and dental wear	Oat (1), Sorghum (3), Rye (2)	
	MI CG F 19	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Oat (1), Spelt (1), Sorghum (2), Barley (2)	
	MI CG F 20	XVII c. AD	ND	Adult ND	ND	Dental calculus and dental wear	Oat (2), Chickpea (1), Broad bean (2), Lentil (1), Sorghum (3), Barley (4), Common wheat (1)	
	MI CG L 21	XVII c. AD	ND	13–19 y	ND	Dental calculus and dental wear	Bean (4), Spelt (3), Barley (2), Common wheat (1), Pea (1)	
	MI CG L 22	XVII c. AD	Female	Adult ND	ND	Partial edentulism, dental calculus and dental wear	Oat (1), Sorghum (1), Barley (1), Corn (1)	
	MI CG L 23	XVII c. AD	ND	Adult ND	ND	Dental calculus, dental wear, dental hypoplasia and caries	Oat (1), Chickpea (1), Broad Bean (1), Lentil (3), Sorghum (1), Barley (15), Rye (5), Common wheat (4), Corn (1)	Horsetail (1)
	MI CG P 24	XVII c. AD	ND	Adult ND	ND	Dental calculus and dental wear	Lentil (1), Sorghum (1), Rye (4), Common wheat (1)	
	MI CG P 25	XVII c. AD	Female	Adult ND	ND	Dental calculus and dental wear	Chickpea (4), Broad bean (2), Lentil (1), Sorghum (2), Barley (4), Rye (3), Millet (1)	
	MI CG P 26	XVII c. AD	Male	Adult ND	ND	Dental calculus and dental wear	Oat (1), Spelt (1), Lentil (4), Barley (2), Buck wheat (1)	
	MI CG P 27	XVII c. AD	Female	Adult ND	ND	Partial edentulism, dental calculus, dental wear and caries	Barley (1), Rye (1)	
	MI CG P 28	XVII c. AD	Male	Adult ND	ND	Dental calculus, dental wear and caries	Barley (2), Rye (2), Common wheat (1)	Horsetail (1)
	MI CG P 29	XVII c. AD	ND	Adult ND	ND	Dental calculus and dental wear	Bean (1), Spelt (1), Broad bean (1), Barley (1), Rye (1), Common wheat (1)	
	MI CG Q 30	XVII c. AD	Male	Adult ND	ND	Dental calculus, dental hypoplasia and caries	Bean (1), Spelt (1), Broad bean (9), Lentil (3), Barley (2), Rye (1)	
	MI CG Q 31	XVII c. AD	Male	Adult ND	ND	Dental calculus	Bean (1), Spelt (1), Broad bean (2), Sorghum (1), Common wheat (1)	
	MI CG Q 32	XVII c. AD	ND	13–19 y	ND	Dental calculus	Sorghum (2), Durum wheat (2)	
	MI CG O 701	XVII c. AD	Male	20–30 y	ND	Dental calculus, dental wear, cribra orbitalia, periostitis		
	MI CG O 702	XVII c. AD	Male	Adult ND	ND	Dental wear and caries		
	MI CG O 703	XVII c. AD	Female	Adult ND	ND	Caries, abscess, and arthritis		
	MI CG O 704		Female		ND	Dental wear and caries		

(continued on next page)

Table 3 (continued)

SITE	ID	Time period	SEX	AGE	Stature	Pathology	Plant remains	Other
		XVII c. AD		Adult ND				
	MI CG O 705	XVII c. AD	Male	13–19 y	ND	Cribra orbitalia		
	MI CG O 706	XVII c. AD	Male	20–30 y	ND	Partial edentulism, caries and syphilis		
	MI CG O 707	XVII c. AD	Male	Adult ND	ND	Partial edentulism and cribra orbitalia		
	MI CG O 708	XVII c. AD	Female	Adult ND	ND	Partial edentulism, cribra orbitalia and caries		
	MI CG O 833	XVII c. AD	Male	Adult ND	ND	ND		
	MI CG O 835	XVII c. AD	Male	Adult ND	ND	ND		
	MI CG O 851	XVII c. AD	Male	20–30 y	ND	ND		
	MI CG O 860	XVII c. AD	ND	Adult ND	ND	ND		
	MI CG O 891	XVII c. AD	Female	Adult ND	ND	Trauma and arthritis		
	MI CG O 894	XVII c. AD	Female	Adult ND	ND	ND		
	MI CG O 895	XVII c. AD	Male	ND	ND	ND		
	MI CG O 1014	XVII c. AD	ND	13–19 y	ND	Cribra orbitalia and caries		
	MI CG O 1143	XVII c. AD	Female	Adult ND	ND	ND		
	MI CG O 1216	XVII c. AD	Female	20–30 y	ND	ND		

Table 4

Number of identified starch grains and their totals in the three sites of the study.

Starch Grains	Milano Ca' Granda	Milano Sabotino	Milano Cattolica	Total
Spelt	34	32	14	80
Barley	89	57	30	176
Rye	64	57	2	123
Corn	10	5		15
Rice	10	5		15
Durum wheat	3			3
Common wheat	26	9	7	42
Oat	24	11	20	55
Millet	6	1	5	12
Sorghum	32	61	11	104
Lentil	41	54	10	105
Broad bean	38	16	1	55
Chickpea	14	11	1	26
Pea	2	1	2	5
Bean	18	15		33
Chestnut	1		1	2
Potato	14			14
Buck wheat	4	2		6
Total	430	337	116	883

populations, although most studies (e.g. Belcastro et al., 2007; Radini et al., 2016) did not focus on the evolution of dietary habits in a specific geographical area throughout time. For the first time, this pilot study presents a comparison based on the analysis of the starch grains trapped in the dental calculus in three archaeological populations from two different historical periods, but from the same geographical area, that is, the city of Milano.

The three archaeological sites analysed in this paper belong to the ancient and modern urban contexts of Milano. The Roman necropolis of Milano Cattolica dates back to the 3rd–4th century CE and consists of interments of individuals from different strata of the society (Sannazaro, 2001). At that time, the city of Milano saw an evolution of its administrative and social situation: from a provincial mercantile city, it was

preparing to become a nerve centre of northern Italy and, at the end of the 3rd century CE, a capital of the Roman Empire (Glaz et al., 2002). The Cattolica necropolis however encompasses many poor burials with few rich ones. The two modern sites of Milano Sabotino and Milano Ca' Granda date back to the 17th century CE and, according to the archaeological and historical records, contained the remains of people from the lower social class. Milano Sabotino represents a common grave where victims of the plague epidemic were buried in the first half of the 17th century (Caruso et al., 2013), whereas the underground chambers of the Milano Ca' Granda contain the skeletal remains of the patients of the city hospital dedicated to the poor (Cosmacini, 1999).

The historical documentation was investigated to find information on dietary habits that could support or contrast our observations. In the Roman period, *puls* – a sort of porridge with millet, rye and barley – and bread made of oat and wheat bran, and vegetables such as olives and chickpeas were the main sources of sustenance for the working class and the poor (Cato, 1954). In Roman times, millet represented a safer and more resistant cultivation than the more delicate wheat and, as pointed out by several ancient authors (i.e. Strabo, 1917), more suitable for moist soils rich in watercourses such as the territory of the city of Milano (Murphy, 2016). Historical sources concerning modern sites report that two kinds of bread were available: the first one was a white bread, composed mainly of white wheat and intended for the aristocrats and clergy, whereas the other was a brown bread made of rye, wheat and millet. This brown bread was only baked in city bakeries and consumed by the lower class (Atti di Governo - Annona p.a., b. 39). After the famine occurred in 1624, the poorest citizens were allowed to bake it in their own homes. According to the records of the city hospital, the hospital did charity works and distributed this bread to the poor, along with legumes, wine and fish, occasionally. Furthermore, patients were also given small portions of dairy products and meat.

In Milano Cattolica, major and minor cereals were equally represented, with a small percentage of legumes. In Milano Sabotino and Milano Ca' Granda, the results showed a higher rate of major cereals, followed by legumes and a lower percentage of minor cereals. The identified starch granules showed a different trend of represented classes

Starch Result

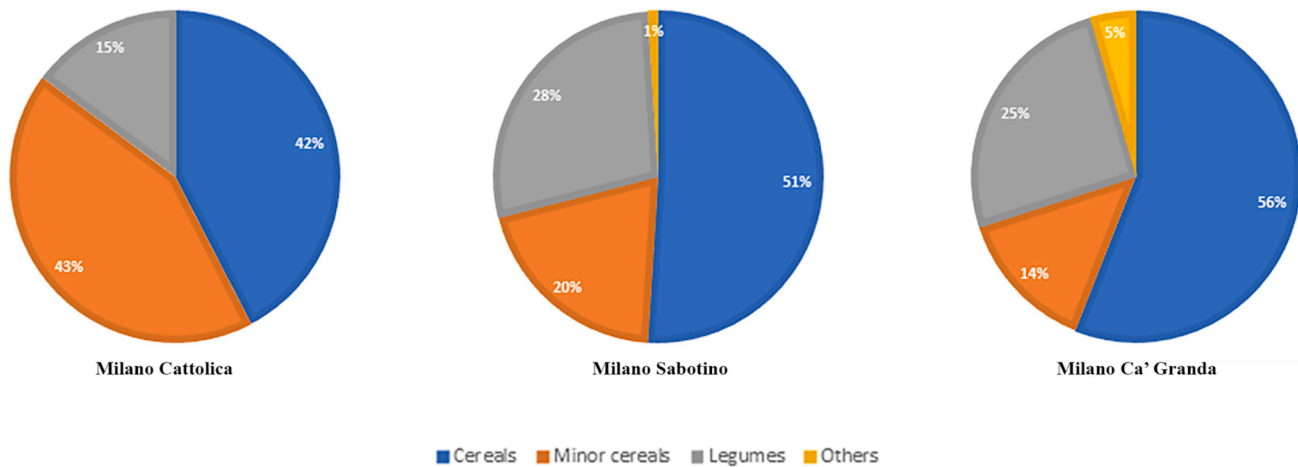


Fig. 3. Starch results in the three sites examined.

throughout time: if in the Roman age legumes are the least represented class, in the modern times legumes seem to replace minor cereals. Overall, major cereals remain the most represented class throughout the time periods considered in this study. The different species observed in the dental calculus of the study sample seem consistent with the reports from historical sources (Castelvetto, 1614; Faas and Whiteside, 2005). Indeed, our findings on the decrease of the minor cereals can be explained with the introduction of various plant species, mainly from America, into the crops of the time. In particular, the presence of corn may explain the lesser cultivation of minor cereals for its high productivity; moreover, the Lombard territory is well suited to respond to the needs of this cereal (Cazzola, 1991). In fact, corn was already cultivated in northern Italy in the second half of the 1500 s, especially in small vegetable gardens (Cazzola, 1991). Within a few decades, it began to expand in the Po Valley and replaced millet and sorghum that were used as animal feed and by poor social classes in times of famine (Cazzola, 1991). Given the similarity to these plants, the cultivation of corn was often opposed because it was thought to cause soil impoverishment but, despite numerous interruptions and given the numerous famines of the 17th and 18th centuries, it became the main food consumed by the poor (Sella, 1980). This intensive consumption also led to the appearance of pellagra which became endemic in northern Italy and was only eradicated in the early 1900s (Finzi, 1982).

The hypothesis of a major diet change in modern times due to the introduction of corn is not supported by the low occurrence of corn starch in our samples. However, corn starch is not easily distinguishable from that of sorghum, which represents the commonest starch type in Milano Sabotino site and is well represented in all the investigated sites. The shift from sorghum to corn could thus have been overlooked. In any case, sorghum has surely represented an important component of diet, given its high frequency also in Milano Cattolica where corn was certainly absent.

On the other hand, the low occurrence of millet is striking and contrasts with the historical sources. Even if small starch grains of millet can be confused with those of rice, even merging the two categories the overall abundance remains low. The small-sized starch grains of millet could have been degraded, or damaged and lost during sample preparation in a higher proportion than the bigger ones from other species. However, even in small absolute quantity, millet has a proportionally higher importance in Milano Cattolica, coherently with the historical sources.

The massive use of legumes could be explained by their easy cultivation even in small plots of land, home gardens or even in pots. Since the Roman age, chickpea lentils and a species of bean (*Vigna unguiculata*) were known and used in northern Italy, but modern American beans (*Phaseolus vulgaris*) were introduced as early as 1534 (Perales, 2001). Although some overlapping of shape and dimension among the legume species could affect the identification at species level, the dominance of lentil and broad bean seems to represent a constant feature throughout the investigated time period.

Finally, little is known on the introduction of the potato in Northern Italy. It is known that the physicist Alessandro Volta brought some specimens from France to a noblewoman of Como (Italy) in 1777 (Visconti, 2013). Only ten years later, in 1785, the government in place decided to obtain seeds from London for cultivation in the Lombard heaths. After numerous experts (including the manager of the botanical garden in Milan) approved this new crop, its cultivation began (Visconti, 2013). The occurrence of potato starch in Ca' Granda samples supports the extension of the time period of the investigated remains to the early-mid 19th century, date at which the chambers of the crypt were reopened to house the bodies of the fallen in the *Risorgimento* battles of 1848 (*Cinque Giornate di Milano*).

Not all starch granules could be successfully identified because of their degraded morphology or because of the overlapping size and shape between plant species. Even if the grouping into major categories helped to address this problem, some misinterpretation could have occurred, for example between different macro categories, as in the sorghum/corn and millet/rice cases discussed above. This issue reiterates the limitation of the starch granules identification based solely on their morphology and dimension, as already reported by Radini et al. (2017).

Another important issue concerns the effect of cooking on grain size and shape. Positive identification occurred through the comparison with raw material, while almost no grain was identified through the comparison with boiled material. The effects of high temperature are very variable within the same sample and make the grains much less visible and identifiable. Probably the grains that can be observed and identified represent a fraction of those less affected by cooking procedures.

The analysis of dental and osseous markers related to health status showed a higher frequency of enamel hypoplasia in Milano Sabotino. Given the similar diet pattern observed in the two modern sites, we can infer that this higher number can be related to the physiological stress caused by the long period of famine that the people buried at Milano

Sabotino endured (Ripamonti et al., 1641). A lower rate of dental stress markers was observed in the Milano Ca' Granda sample and this could be explained by the fact that the hospital undertook again charity works, providing sustenance to the lower class in the second half of the 17th century (especially abandoned children, Cosmacini, 1999). However, there was no strong evidence suggesting the presence of food-related anemias. *Cribra orbitalia* is a non-specific sign and its presence may be due to various pathological factors (e.g. neoplastic, infectious and traumatic lesions) (Wapler, Crubézy and Schultz, 2004; Oxenham and Cavill, 2010; Rivera and Mirazón Lahr, 2017; Buikstra, 2019b). Finally, no skeletal evidence of specific nutritional deficiencies was found, specifically, vitamin C and D. Consequently, despite the variety of the diet extrapolated from the analysis of dental calculus, which included both major and minor cereals as well as vegetables, the individuals of all three populations (and to a higher degree, that of Milano Sabotino) experienced physiological stress as evidenced by the presence of enamel hypoplasia and *cribra orbitalia*, which is consistent with our knowledge of the time.

Not all of the material extracted from dental calculus represents plants known for food consumption. In our case, the most significant findings are the horsetail spores (*Equisetum* cfr. *Arvense*, Fig. 4). According to the 17th century pharmacopoeia, horsetail was used for medical purposes as a diuretic and clotting agent, and the sterile stems served for oral hygiene and dish cleaning (Ca' Granda's archival source). Evidence supporting the medicinal use of this plant includes a pharmaceutical container from the Milano Ca' Granda collection of hospital vases with the words "*Aqua aequiseti*" (which means "equisetum water") written on it. However, it is difficult to explain the presence of these traces in dental calculus from their medicinal use or as remains of herbal teas and decoctions: furthermore, these uses involve mainly the more available sterile stems, while the occurrence of spores involves necessarily the use of fertile ones. Food use of fertile stems has been reported in Europe (Turner et al., 2011; Luczaj et al., 2012) and even nowadays, horsetail is used for its beneficial properties. This theory cannot be excluded, being in this case a useful indicator of poor living conditions. Still, little is known about the actual use of this plant and how it can be entrapped in dental calculus (EFSA Panel on Dietetic Products, 2009).

Further studies include an increase in the sample size as well as specimens from different time periods and other archaeological sites from the city of Milano. In addition, the analysis of dental calculus could be enhanced by other investigations (e.g., trace elements) which would allow to better understand the balance of the foods consumed and the presence of animal proteins.

5. Conclusion

This study represents a starting point to better understand the dietary habits of the ancient citizens of the city of Milano. Anthropological and starch analyses were carried out on dental calculus from a subset of 55 individuals of 150 total skeletons from three archaeological sites of Milano dated to the Roman and Modern eras. This diachronic approach showed that major cereals were a constant finding over the two time periods considered, whereas minor cereals appear to have been replaced by legumes in the Modern Age. The results of our study are consistent with Roman and Modern Age archival and historical accounts and other studies on coeval populations. The analyses of stress markers did not reveal severe nutritional deficiencies or signs of poor diet in the sample tested.

CRediT authorship contribution statement

Mirko Mattia: Conceptualization, Writing – original draft, Investigation. **Lucie Biehler-Gomez:** Writing – original draft, Formal analysis. **Andrea Palamenghi:** Writing – original draft, Formal analysis. **Deborah Nichetti:** Methodology, Investigation. **Giulia Caccia:** Writing – original draft, Methodology, Investigation. **Emanuela Sguazza:**

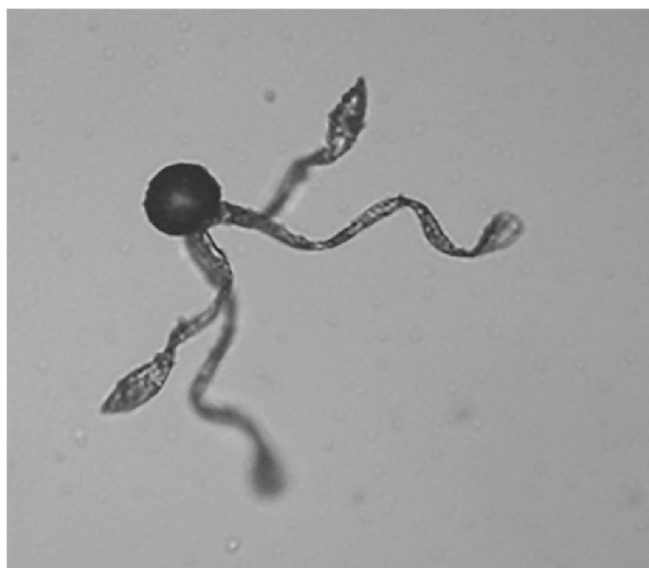


Fig. 4. *Equisetum arvense* spores.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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